

Concept Summary: Removable Bulkhead Utility Interface System (RBUIS)

Overview

The Removable Bulkhead Utility Interface System (RBUIS) is a modular utility transfer assembly designed to provide electrical power and network communications between two adjacent structures through a removable bulkhead installed on existing door-pin flange interfaces. The system consists of two physically independent connection plates: a Power Interface Plate Assembly (PIPA) and a Data Interface Plate Assembly (DIPA).

The design objective is to enable rapid installation and removal of the bulkhead while maintaining safety, reliability, environmental protection, and compliance with applicable electrical, building, and telecommunications standards.

System Architecture

Utility Separation

The bulkhead incorporates two independent utility interfaces:

1. Power Interface Plate Assembly (PIPA)

- Electrical power transfer only.
- No communications conductors.
- Fully interlocked and protected.

2. Data Interface Plate Assembly (DIPA)

- Fiber-optic communications only.
- No metallic communication conductors.
- Electrically isolated from power system.

The assemblies are mechanically independent and may be serviced or upgraded separately.

Power Interface Plate Assembly (PIPA)

Functional Description

The PIPA provides blind-mate electrical power transfer between the source structure and the receiving structure through a removable docking interface.

Contact Orientation

Supply Side

- Male contact assembly.

- Recessed behind safety barriers and interlocked access shutters.
- Mounted within the source-side bulkhead frame.

Demand Side

- Female contact assembly.
- Finger-safe recessed receptacles.
- Mounted within the receiving-side bulkhead frame.

This arrangement ensures that energized conductors remain on the protected supply side and that the receiving side presents no exposed conductive elements during disconnect conditions.

Power Connection Sequence

1. Bulkhead installed and mechanically secured.
2. Alignment guides engage.
3. Grounding contacts engage first.
4. Power contacts fully seat.
5. Mechanical lock verified.
6. Electrical interlocks verified.
7. Source contactor closes.
8. Power transfer enabled.

Disconnect Sequence

1. Source contactor opens.
2. Zero-energy state confirmed.
3. Mechanical lock released.
4. Power contacts separate.
5. Ground contacts separate last.

Key Features

- Blind-mate operation.
- Ground-first / ground-last engagement.
- Silver-plated copper contacts.
- Arc-resistant contact chambers.
- Mechanical and electrical interlocks.
- Environmental sealing.
- Replaceable contact modules.
- Lockout/tagout capability.

Data Interface Plate Assembly (DIPA)

Functional Description

The DIPA provides high-speed communications between structures through a dedicated fiber-optic connection system integrated into the removable bulkhead.

Interface Design

Structure A

- MPO/MTP fiber cassette assembly.
- Spring-mounted alignment system.
- Dust-protected optical shutters.

Structure B

- Matching MPO/MTP fiber cassette assembly.
- Blind-mate optical alignment system.
- Environmental sealing.

Communications Medium

- Single-mode fiber optic backbone.
- No copper communication conductors crossing between structures.
- No shared communication grounding path.

Key Features

- Blind-mate fiber connection.
 - Multi-fiber backbone architecture.
 - Dual redundant optical paths.
 - Expansion fibers for future use.
 - High-bandwidth capability.
 - Lightning immunity.
 - Ground-potential isolation.
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Bulkhead Assembly

Mechanical Interface

The bulkhead is designed as a removable panel assembly that attaches directly to existing door-pin flange interfaces.

Structural Requirements

- Utility plates are non-structural.
- Bulkhead frame carries mounting loads.
- Utility plates mounted on floating isolation brackets.

- Differential movement accommodation incorporated.

Environmental Requirements

- Weather-resistant construction.
 - Exterior-rated seals and gaskets.
 - Corrosion-resistant materials.
 - Dust and water ingress protection.
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Safety Philosophy

The design follows four fundamental principles:

1. De-Energized Separation

Power conductors cannot be separated while energized.

2. Protected Source Contacts

Male power contacts are located on the supply side and remain inaccessible unless all interlocks are satisfied.

3. Fiber-Based Communications

No metallic data conductors pass between structures, eliminating communication-related grounding and surge concerns.

4. Independent Utility Systems

Power and communications systems remain physically and electrically isolated.

Intended End State

When installed, the removable bulkhead functions as a self-contained utility docking module that:

- Transfers electrical power from a source structure to a receiving structure.
- Transfers network communications through isolated fiber-optic links.
- Attaches using existing door-pin flange infrastructure.
- Can be removed without exposing energized conductors.
- Maintains separation of power and communications systems.
- Supports inspection, maintenance, and future expansion without modification to the host structures.

The resulting system provides a repeatable, code-compliant utility interface suitable for modular facilities, relocatable structures, mobile infrastructure, temporary facilities, and rapidly deployable building systems.



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